SESSION 4 — MORNING

Corporate Finance

Capital budgeting, WACC, capital structure & leverage

Learning Objectives

Capital Budgeting

Apply NPV, IRR, payback and PI decision rules. Know which method to use and when they conflict.

WACC

Calculate the weighted average cost of capital from first principles. Apply CAPM for cost of equity.

Capital Structure

Understand M&M theorem (with and without taxes). Apply the trade-off between tax shield and distress costs.

Operating & Financial Leverage

Calculate DOL, DFL and DTL. Understand their impact on earnings volatility and risk.

Capital Budgeting — The Investment Decision Framework

Capital budgeting is the process of evaluating and selecting long-term investments. These decisions shape the firm's future.

NPV	★★★★★	Preferred method. Directly measures value created in £. Accounts for time value, risk and opportunity cost. Always choose highest NPV.	Accept if NPV > 0
IRR	★★★★☆	Intuitively appealing. Conflicts with NPV for mutually exclusive projects (scale problem). Use as secondary check.	Accept if IRR > hurdle rate
Payback	★★☆☆☆	Simple and intuitive. Ignores time value of money and all cash flows beyond payback. Useful as quick liquidity screen only.	Accept if payback < target
Profitability Index	★★★☆☆	PI = PV of inflows / Investment. Useful when capital is rationed — ranks projects by value per £ invested.	Accept if PI > 1

WACC — The Corporate Hurdle Rate

$$WACC = (E/V) \times r_e + (D/V) \times rd \times (1 - t)$$

1

Capital structure weights

E/V = Equity/(Equity+Debt). D/V = Debt/(Equity+Debt). Use MARKET values, not book values.

2

Cost of debt (rd)

Yield on existing debt or marginal cost of new debt. Adjust for tax: after-tax cost = rd × (1−t).

3

Cost of equity (r_e) via CAPM

$r_e = Rf + \beta \times (Rm - Rf)$. Rf = risk-free rate (10yr Gilt). β = systematic risk. (Rm−Rf) = equity risk premium (~5-6%).

4

Blend

WACC = weighted average. Represents the minimum return the firm must earn on investments to create value.

WORKED EXAMPLE

Firm: E = £600m, D = £400m (total V = £1,000m). rd = 6%, Tax = 25%, Rf = 4%, β = 1.2, ERP = 5%.
r_e = 4 + 1.2×5 = 10%. After-tax rd = 6%×0.75 = 4.5%.
WACC = (0.6×10%) + (0.4×4.5%) = 6.0% + 1.8% = 7.8%

Modigliani-Miller — The Foundation of Capital Structure

Without taxes, capital structure is irrelevant. Value = PV of operating cash flows, not how you finance them.

M&M I (No Tax)	V(Levered) = V(Unlevered)	Pie analogy: slicing a pie differently doesn't change its size. Debt and equity just split the same cash flows.	*Leverage doesn't create value on its own.*
M&M I (With Tax)	V(L) = V(U) + PV(Tax Shield)	Interest is tax-deductible → debt creates a tax shield = t × D. This suggests 100% debt is optimal (in theory!).	*Tax shield adds value but is bounded by distress costs.*
M&M II	$r_e = r_0 + (D/E)(r_0 - rd)$	Cost of equity rises linearly with leverage. The benefit of cheaper debt is offset by rising equity risk.	*WACC is constant (no tax) — no free lunch.*
Trade-Off Theory	Optimal D/E where V is maximised	Balance tax shield benefits against bankruptcy/distress costs. Optimal leverage is firm-specific.	*Explains why companies don't go 100% debt despite tax shield.*

Operating & Financial Leverage

Leverage amplifies changes. Understand both types to model earnings volatility under different scenarios.

DOL = % Δ EBIT / % Δ Revenue DFL = % Δ EPS / % Δ EBIT DTL = DOL × DFL

Operating Leverage (DOL)

Driven by fixed vs variable cost mix.

High FC → high DOL → revenue changes amplified into larger EBIT swings.

DOL = (Revenue − VC) / EBIT = Contribution / EBIT

Ex: DOL = 3.0 means 10% revenue rise → 30% EBIT rise (and vice versa).

High DOL sectors: Airlines, hotels, car manufacturers.

Financial Leverage (DFL)

Driven by fixed interest obligations.

DFL = EBIT / (EBIT − Interest)

Ex: EBIT=£100m, Interest=£20m → DFL = 100/80 = 1.25× EBIT changes amplified 25% into EPS.

High DFL → higher EPS in good times, higher distress risk in downturns.

High DFL sectors: Private equity-backed companies (LBOs), REITs, utilities.

PRACTICE PROBLEMS

Try before turning the page.

Q1	Project: invest £200,000. Cash flows: Yr1=£60k, Yr2=£75k, Yr3=£80k, Yr4=£70k. WACC=10%. Calculate NPV and IRR (approximate). Accept?
Q2	Firm: Equity=£500m (market), Debt=£300m, rd=5%, Tax=22%, Rf=3.5%, Beta=1.4, ERP=5.5%. Calculate WACC step by step.
Q3	Revenue = £400k, VC = £240k, FC = £80k, Interest = £12k. Calculate EBIT, DOL, DFL and DTL.
Q4	Company A has EBIT=£50m and interest=£10m. Under M&M with 25% tax, what is the annual tax shield benefit from the debt?
Q5	If a company increases financial leverage, what happens to: (a) WACC, (b) cost of equity, (c) cost of debt per M&M Prop II?

Practice — Answers

A1

PV inflows = 54,545+61,983+60,105+47,811 = £224,444. NPV = 224,444−200,000 = +£24,444 → ACCEPT. IRR ≈ 15-16% (trial: at 15% NPV≈+£8k, at 17% NPV≈−£2k → IRR ≈ 16.5%).

A2

r_e = 3.5+1.4×5.5 = 11.2%. After-tax rd = 5%×0.78 = 3.9%. E/V=500/800=62.5%, D/V=37.5%. WACC = 0.625×11.2 + 0.375×3.9 = 7.0+1.46 = 8.46%.

A3

Contribution = 400−240 = £160k. EBIT = 160−80 = £80k. DOL = 160/80 = 2.0×. DFL = 80/(80−12) = 80/68 = 1.18×. DTL = 2.0×1.18 = 2.35×.

A4

Annual tax shield = Tax rate × Interest = 25% × (Debt × rd). If rd=6%, D=200m: Interest=12m → Tax shield = 0.25×12m = £3m/yr. PV(tax shield) = 3m/0.06 = £50m (assuming perpetual debt).

A5

(a) WACC: under M&M no tax = unchanged (WACC constant). With taxes = falls (tax shield benefit). (b) Cost of equity: rises (more financial risk per M&M Prop II). (c) Cost of debt: unchanged (debtholders' risk doesn't change).

KEY TAKEAWAYS

Session 7 — Corporate Finance

- NPV is the gold standard — always choose highest NPV for value creation
- WACC = blended cost of debt+equity = corporate hurdle rate
- $r_e = Rf + \beta \times ERP$ — risk-free plus reward for systematic risk
- M&M: tax shield adds value; distress costs set the optimal leverage limit
- DOL amplifies EBIT from revenue; DFL amplifies EPS from EBIT
- Trade-off theory: optimal D/E where tax shield PV = PV distress costs